

Design by Contract

If the caller meets the requirements, the routine must deliver its promises.

THE AGREEMENT

EXAMPLE: ADD A CHILD TO A TREE NODE

Precondition	Caller must satisfy it before the call.	<code>new</code> refers to an existing node.
Postcondition	Routine guarantees it on successful return.*	<code>new.parent = current</code> <code>child count = old child count + 1</code>
Class invariant	Class stays consistent at public boundaries.	Each existing child points back to its parent node.

*Given valid entry conditions. Invariants hold after creation and at public operation entry/exit, not every internal step.

Subclasses: demand no more, promise no less; preserve inherited invariants.

Failure: restore consistency, then retry or report failure — never silently claim success.